

Assignment 3: AI-Based Requirement Refinement using Clarification Questions

Problem Statement

Stakeholder requirements are often incomplete, ambiguous, or vague, which can lead to unclear Functional Requirements (FR) and Non-Functional Requirements (NFR). Simply extracting requirements from recordings or emails may produce inaccurate or missing information.

Develop an AI-powered system that not only extracts requirements from recordings/emails but also identifies ambiguities and asks clarification questions to refine the requirements before generating the final Software Requirement Specification.

The system should interact intelligently with the user to improve requirement quality and completeness.

Objective

The objective of this assignment is to:

- Understand ambiguity in Requirement Engineering
 - Use AI to identify incomplete or unclear requirements
 - Generate clarification questions automatically
 - Refine Functional and Non-Functional Requirements through interaction
 - Improve requirement quality using iterative feedback
 - Generate structured and refined requirement specifications
-

Features to Implement

1. Upload audio/video recordings or fetch emails
2. Convert recordings into transcripts (if applicable)
3. Extract initial Functional Requirements (FR)
4. Extract initial Non-Functional Requirements (NFR)
5. Detect vague, ambiguous, or incomplete requirements
6. Automatically generate clarification questions
7. Accept user/stakeholder responses for clarification
8. Refine and update FRs and NFRs based on responses
9. Categorize NFRs (Security, Performance, Reliability, etc.)
10. Display both initial and refined requirements

11. Export final refined requirements as PDF/DOCX/TXT